4.2 Harris' Corner Detector

Harris' corner detector is an improvement over Moravec's corner detector. It uses a more sophisticated error function and calculates the cornerness of each pixel in an image. The main idea is to detect regions in the image where intensity changes sharply in all directions — corners.

4.2.1 Structure Matrix

Harris' corner detector uses an error function that is formulated as the **continuous version of Moravec's detector**. The error function is derived by considering an **infinitesimal shift** $(\Delta x, \Delta y)$ of the image:

$$E(\Delta x, \Delta y) = \sum_{x,y} w(x,y) \left(I(x+\Delta x, y+\Delta y) - I(x,y)\right)^2$$

Where:

- $w(x,y)$ is the **window function** (mask) that determines the region of the image over which we compute the error.
- $I(x,y)$ represents the image intensity at pixel (x,y).

Taylor Series Expansion

Using the **Taylor series expansion** for the intensity difference, we approximate the intensity difference as:

$$I(x+\Delta x, y+\Delta y) - I(x,y) \approx \partial_x I(x,y)\Delta x + \partial_y I(x,y)\Delta y$$

Where $\partial_x I(x,y)$ and $\partial_y I(x,y)$ are the **image gradients** in the x and y directions, respectively.

Matrix Formulation

The error function can be rewritten in a matrix form, resulting in a **structure matrix** M_w:

$$M_w = \begin{bmatrix} \sum_{x,y} w(x,y)(\partial_x I(x,y))^2 & \sum_{x,y} w(x,y)\partial_x I(x,y)\partial_y I(x,y) \\ \sum_{x,y} w(x,y)\partial_x I(x,y)\partial_y I(x,y) & \sum_{x,y} w(x,y)(\partial_y I(x,y))^2 \end{bmatrix}$$

This matrix encodes the **local structure of the image** at each pixel. It is **real and symmetric**, so it can be diagonalized by an orthogonal matrix R, and its eigenvalues λ_1 and λ_2 describe the intensity variation in different directions.

Eigenvalues of the Structure Matrix

The eigenvalues λ_1 and λ_2 of the structure matrix represent the change in intensity along the directions of the principal eigenvectors. The relationship between these eigenvalues helps classify regions of the image:

- **Flat Region**: Both eigenvalues λ_1, λ_2 are small, indicating minimal intensity change in all directions.
- **Edge**: One eigenvalue is large, and the other is small, indicating intensity change in one direction (edge).
- **Corner**: Both eigenvalues are large, indicating intensity change in all directions.

Error Function and Eigenvalues

The error function can be expressed in terms of the eigenvalues:

$$E(\Delta x, \Delta y) = \Delta x \Delta y \begin{bmatrix} \lambda_1 & \lambda_1 \partial_x I \partial_y I \\ \lambda_1 \partial_x I \partial_y I & \lambda_2 \end{bmatrix} \Delta x \Delta y$$

Where:

- λ_1 and λ_2 control the shift direction and the intensity change at each pixel.

4.2.2 Harris Cornerness Function

The **cornerness** at each pixel is computed from the eigenvalues of the structure matrix M_w. A more efficie approximation for cornerness is given by:

$$C(x, y) = \lambda_1 \lambda_2 - k(\lambda_1 + \lambda_2)^2$$

Where:

- λ_1, λ_2 are the eigenvalues of M_w.
- k is a constant (empirically set in the range of 0.04 to 0.06).

This cornerness function is used to evaluate each pixel in the image.

Cornerness Evaluation:

- **Flat Region**: $C(x, y)$ is small.
- **Edge**: $C(x, y)$ is small.
- **Corner**: $C(x, y)$ is large.

After calculating the cornerness for each pixel, **thresholding** and **non-maximum suppression (NMS)** are applied to identify the actual corner points.

4.2.3 Properties of Harris' Corner Detector

Harris' corner detector has several important properties:

- **Rotation Invariance**: The eigenvalues λ1\lambda_1λ1 and λ2\lambda_2λ2 are invariant under image rotation, making the detector rotation-invariant.
- **No Affine Intensity Change Invariance**: The detector is invariant to affine intensity changes (e.g., scaling, translation, and rotation) but **not invariant to a multiplicative change in intensity** (gain factor).
- **No Scale Invariance**: The detector uses a fixed window size, meaning it is **not scale-invariant**. This limitation can be addressed by using multi-scale techniques.
- **Invariance to Bias**: The detector is invariant to an additive bias (e.g., if the image intensity is shifted by a constant value).

4.2.4 Harris Corner Detection Algorithm

The steps for Harris' corner detection are as follows:

1. **Compute the cornerness** C(x,y)C(x, y)C(x,y) at each pixel using the cornerness function.
2. **Thresholding**: Select pixels where C(x,y)C(x, y)C(x,y) is higher than a chosen threshold TTT.
3. **Non-Maximum Suppression (NMS)**: From the set of thresholded pixels, keep only the local maxima of C(x,y)C(x, y)C(x,y) to identify distinct corner points.